

Classwork (21st July)

1. BRAC University academic support centre wants to investigate how different student behaviours influence performance in the final STA201 examination. Data were collected from 15 students who attended a preparation programme. For each student, the following variables were recorded: study hours per week (X_1), average hours of sleep per night (X_2), class attendance percentage (X_3), number of practice tests completed (X_4), and final exam score out of 100 (Y). The centre wishes to examine the relationships between these variables and determine which factors are useful for predicting exam performance. The collected data are shown below:

Student	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
StudyHours	2	3	4	5	6	7	8	9	10	11	12	13	6	7	9
SleepHours	6	6	7	6	7	6	7	7	8	7	8	7	6	7	8
Attendance	60	65	70	72	75	80	85	88	90	92	94	95	76	82	89
PracticeTest	0	1	1	2	2	3	3	4	4	5	5	6	2	3	4
ExamScore	50	54	58	60	65	70	74	78	82	86	90	92	66	72	80

- (a) Identify the dependent variable and the independent variables in this study and briefly explain why the dependent variable is considered the response variable in this context. (2 marks)
- (b) Using Exam Score (Y) and Study Hours (X_1), draw an appropriate scatterplot and describe the direction, strength, and form of the relationship between the two variables. (3 marks)
- (c) Calculate the Pearson correlation coefficient between Exam Score (Y) and Study Hours (X_1), calculate the coefficient of determination, and interpret both measures in the context of the relationship between study hours and exam performance. (5 marks)
- (d) Fit a simple linear regression model with Exam Score (Y) as the dependent variable and Study Hours (X_1) as the explanatory variable. Write the estimated regression equation, interpret the estimated intercept and slope in the context of the study, and use the fitted model to predict the exam score of a student who studies 10 hours per week. (5 marks)
- (e) Fit another simple linear regression model with Exam Score (Y) as the dependent variable and Attendance (X_3) as the explanatory variable, write the estimated regression equation, calculate the coefficient of determination for this model, and interpret the value obtained. (4 marks)
- (f) Compare the regression models obtained in parts (d) and (e) in terms of their coefficients of determination and determine which model explains more variation in exam scores. Justify your conclusion. (2 marks)

- (g) Draw a graph showing the fitted regression lines from the two models obtained in parts (d) and (e) on the same plot and briefly comment on the differences between the two fitted lines. (2 marks)
- (h) Suppose the following multiple linear regression model was obtained using all explanatory variables Interpret the intercept and each of the estimated regression coefficients in the context of this study. (3 marks)

$$\hat{Y} = 15 + 2.8X_1 + 0.9X_2 + 0.25X_3 + 2.5X_4$$

- (i) Suppose the coefficient of multiple determination for the model in part (h) is $R^2 = 0.92$. Interpret this value, calculate the adjusted R^2 given that there are 15 observations and 4 predictors, and briefly explain why adjusted R^2 is generally preferred to R^2 in multiple regression analysis. (2 marks)
- (j) Consider a regression model in which no independent variables are included. Write the form of this model, fit it using the given data, draw the fitted model, and explain whether meaningful predictions can be made from such a model. (1 mark)
- (k) Using the multiple regression model given in part (h), calculate the predicted exam score for a student with Study Hours = 8, Sleep Hours = 7, Attendance = 85, and Practice Tests = 3, and interpret the predicted value in the context of the study. (1 mark)